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Desigualdad triangular inversa para normas

Proposición 1 (Desigualdad triangular inversa). *Sea* $(X, \|\cdot\|)$ *un espacio normado*

$$\implies \forall x, y \in X : |\|x\| - \|y\|| \leqslant \|x - y\| .$$

***Demostración*:** Por la `metrica-inducida/??`, $\forall x, y \in X : d(x, y) := \|x - y\|$ define una métrica en X, luego por la desigualdad triangular inversa para métricas se cumple que

$$\forall x, y \in X : |d(x, 0) - d(y, 0)| \leqslant d(x, y) \implies |\|x\| - \|y\|| \leqslant \|x - y\| .$$

■

Referenciado en

- `prop-norma-continua`

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